



OneATS

User manual

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TAKAOKA TOKO CO., LTD.



OneATS user manual

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Symbol	Description
CIM	Common Information Model
DMS	Distributed Management System
DCOM	Distributed Component Object Model
FEP	Front End Processing
HMI	Human Machine Interface
I/O	Input/Output
IED	Intelligent Electronic Device
LAN	Local Area Network
OPC UA	OPC Unified Architecture
RTU	Remote Terminal Unit
SCADA	Supervisory Control and Data Acquisition
SSL	Secure Socker Layer
TCP	Transmission Control Protocol
TLS	Transport Layer Security
XML	Extensible Markup Language

Text color:

- Comment
- Notes
- Warnings

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B.	Database
C.	OneATS DataEditor – Project Management
D.	OneATS FEP
E.	OneATS DataServer
F.	OneATS Grid Studio
G.	OneATS SmartHIS
H.	OneMasterTool



OneATS User Manual

- MongoDB

12/Jun/2024

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Software Version	Date of Issue	Change Description
V4.2.5	09/Apr 2020	



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B. MongoDB

B.1. Overview MongoDB in OneATS

B.1.1. MongoDB Architecture

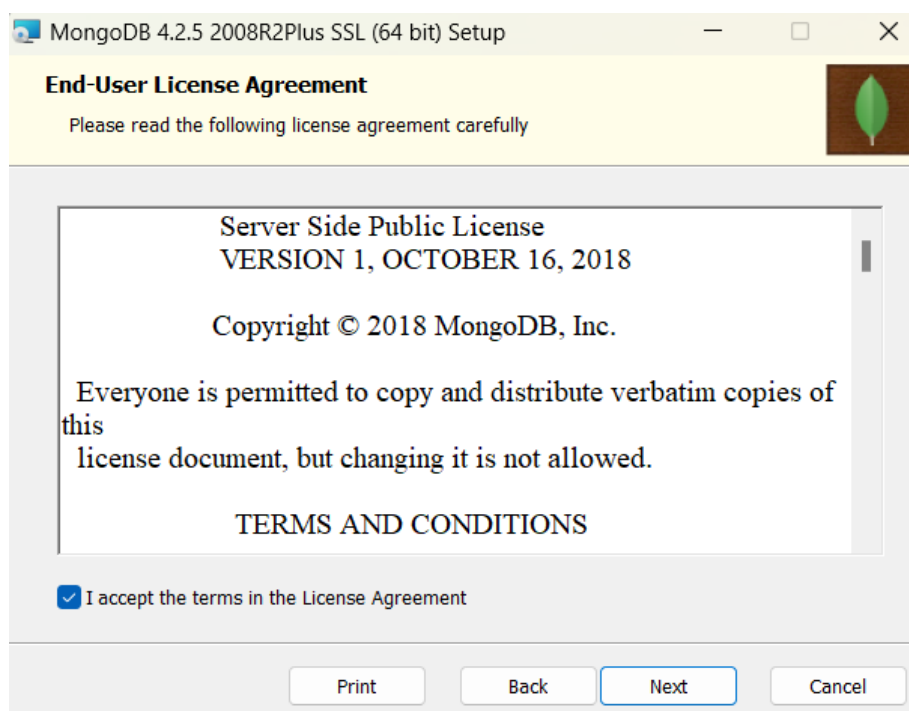
B.2. Install

B.2.1. Install the app

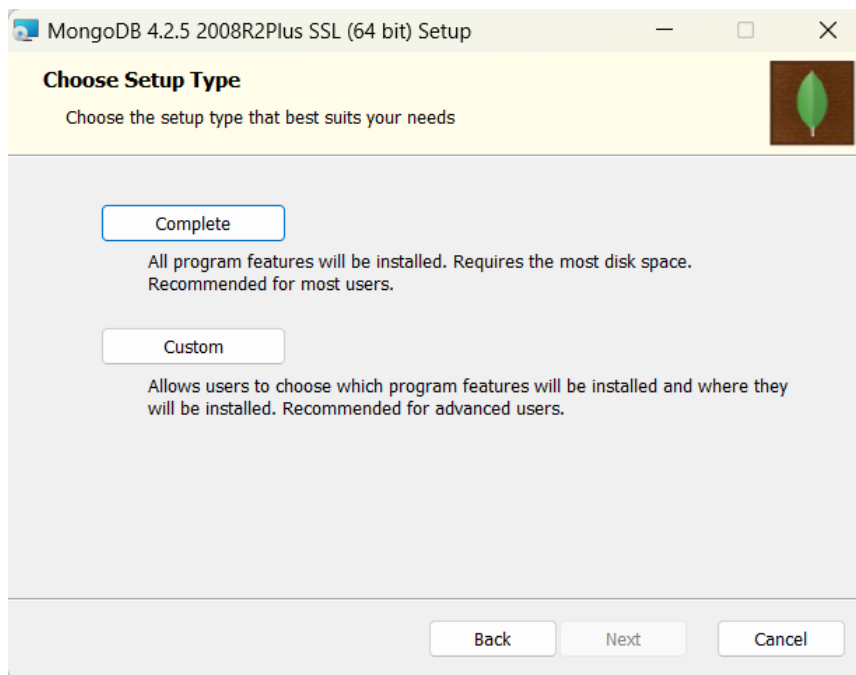
Run the mongodb-win32-x86_64-2012plus-4.2.5-signed.msi installation file. Select **Next** to continue:



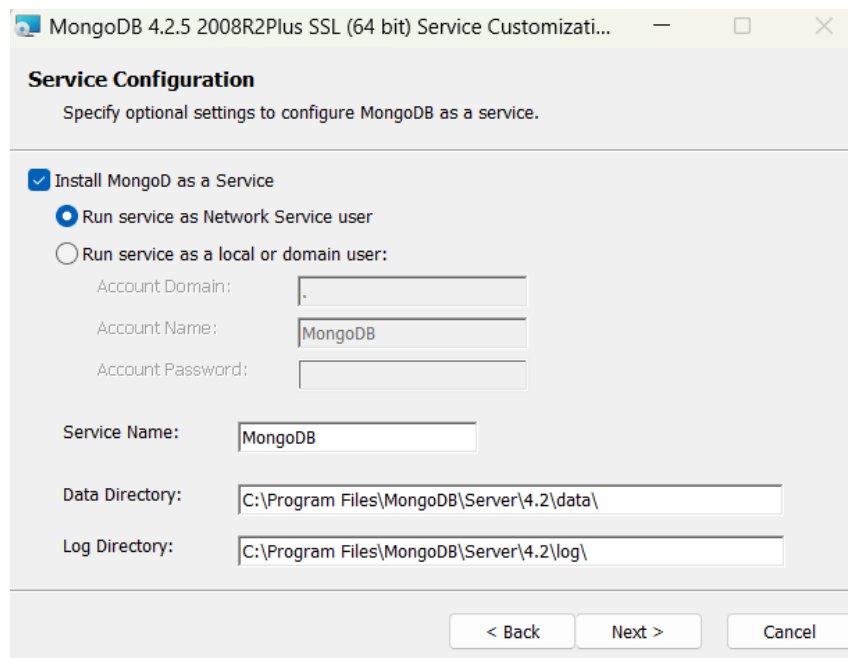
Select **I accept the terms in the License Agreement**, then click **Next**:



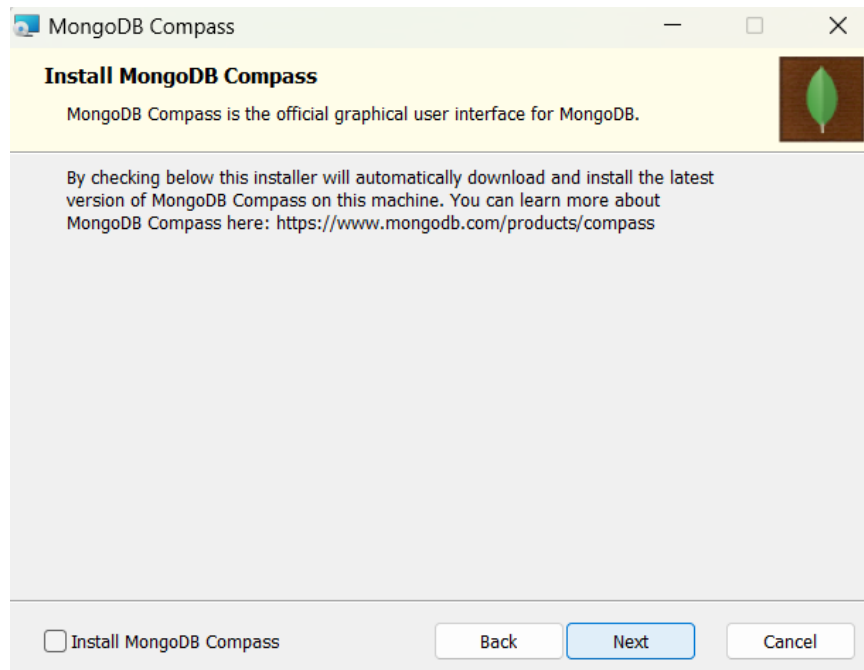
Click **Complete**:



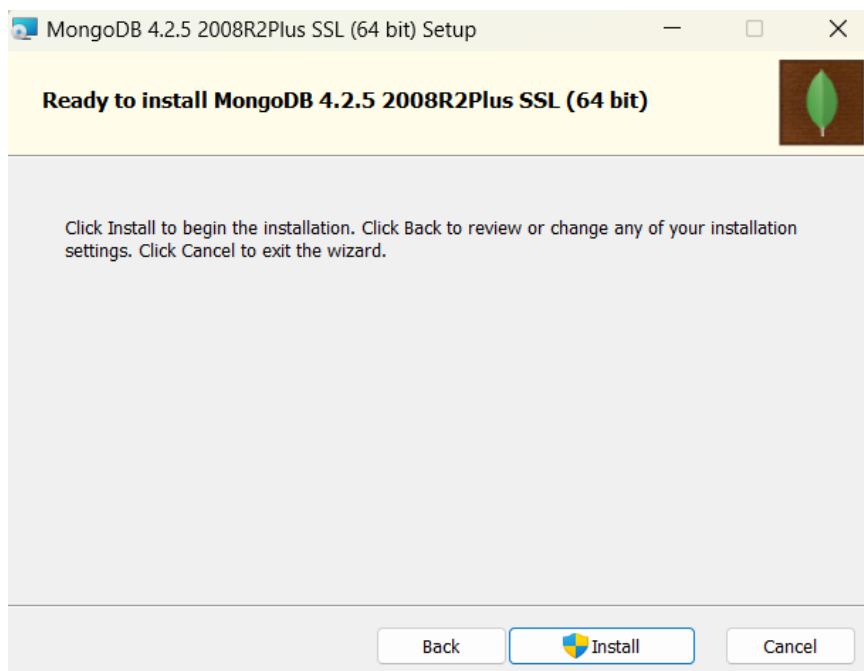
Fill in the path to the database data folder at **Data Directory**, the path to the log directory at **Log Directory** (recommended: choose a different operating system partition). Then click **Next**:



Without selecting **Install MongoDB Compass**, click **Next** to continue:



Click **Install** and wait for the installation process to take place:



Select **Finish** to finish the installation process:

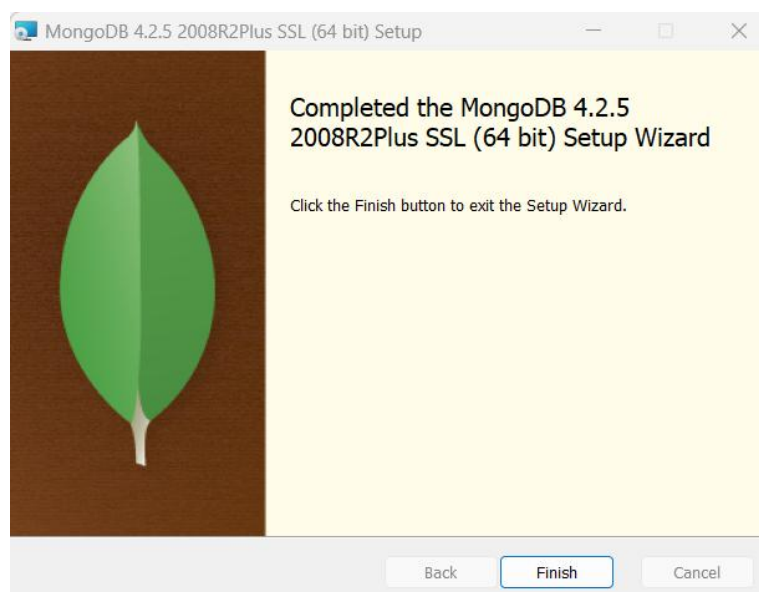


Figure B-1. Finish the MongoDB installation process

After the MongoDB installation, we install *OneATS Data Editor* to serve the Replication deployment.

B.2.2. Configuring MongoDB

B.2.2.1 Initializing MongoDB for OneATS

Stop service **MongoDB Server** in Window Services

	MongoDB Server	MongoDB Database Server	Automatic
	Natural Authentication	Signal aggregator service, that...	Manual (Trigger Start)

Prepare the OneATS database **initialization directory (MongoDB Initiate)**, including:

- The data **folder** contains data that has been created with 1 account:

Default User: Administrator

Default Password: atsco

- The **mongod.cfg file** is a sample mongoDB server configuration file.
- The **oakeyfile.key** file is used to configure **Replication** and is not necessary if only Standalone is configured.

Copy **oakeyfile.key** in the **MongoDB Initiate** folder to:

C:\Program Files\MongoDB \Server\4.2\bin

Modify **the mongod.cfg file** accordingly at the directory address:

C:\Program Files\MongoDB \Server\4.2\bin

```

1 systemLog:
2   destination: file
3   path: C:\Program Files\MongoDB\Server\4.2\log\mongod.log
4 storage:
5   dbPath: C:\Program Files\MongoDB\Server\4.2\data
6 net:
7   bindIp: 0.0.0.0
8   port: 27017
9 security:
10  keyFile: C:\Program Files\MongoDB\Server\4.2\bin\oakeyfile.key
11  authorization: enabled
12 #replication:
13 #   replSetName: "rsOA"

```

Figure B-2. File mongod.cfg

storage dbPath and **systemLog path** are the paths that are declared in the Data Directory and Log Directory respectively when installing MongoDB.

- **net:** bindIp, port – addresses and ports that listen to connections from applications:

#network interfaces

Net:

Port: 27017

bindIp: 127.0.0.1

- **security:** keyFile – the path to file oakeyfile.log:

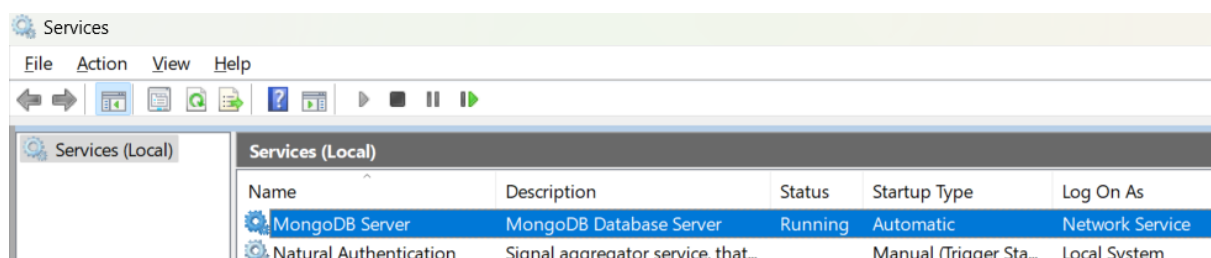
#security

Security:

keyFile: C:\Program Files\MongoDB\Server\4.2\bin\oakeyfile.key
authorization: enabled

Replace all contents in the original **Data** folder during installation (for example, C:\Program Files\MongoDB\Server\4.2\data) with data in the **MongoDB Initiate\data** directory.

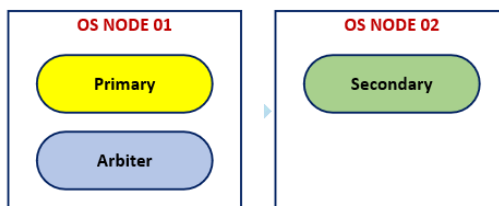
Run the **MongoDB Server** service again in Window Services:



B.2.2.2 Install MongoDB multi-instance on the same operating system node.

In some situations, it is necessary to set up a MongoDB multi-instance on 1 operating system to deploy the Replication feature. Typical scenarios include:

- The system only has 02 operating system nodes. Structure for implementing MongoDB with Replication.



- Set up the Production Environment on the Engineering Computer/ Development Environment. The real system has many MongoDB nodes, on 01 Engineering Computer it is necessary to set up a Production Environment with many MongoDB nodes (for example, to perform tasks to monitor mongoDB nodes).

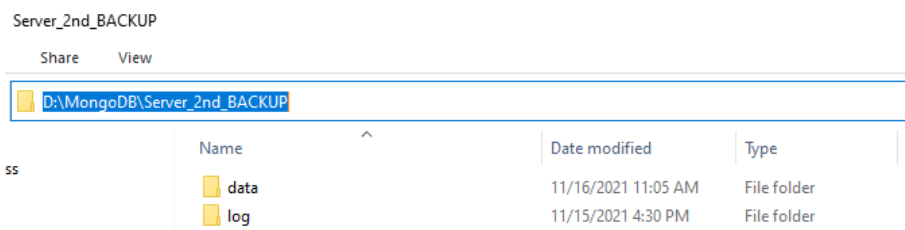
Each MongoDB instance will correspond:

- 01 MongoDB Database folder, including \data and \log subfolders.
- 01 *.cfg configuration file, which stipulates 01 TCP port independent of other services.
- 01 MongoDB service with the corresponding configuration file parameters.

Installation steps for 01 MongoDB instance

Step 1: Create a Database folder.

- Create a MongoDB Database directory by copy-pasting from an existing Database directory. It includes 2 folders \data and \log. (for example, D:\MongoDB\Server_2nd_BACKUP).



Step 2: Create a configuration file.

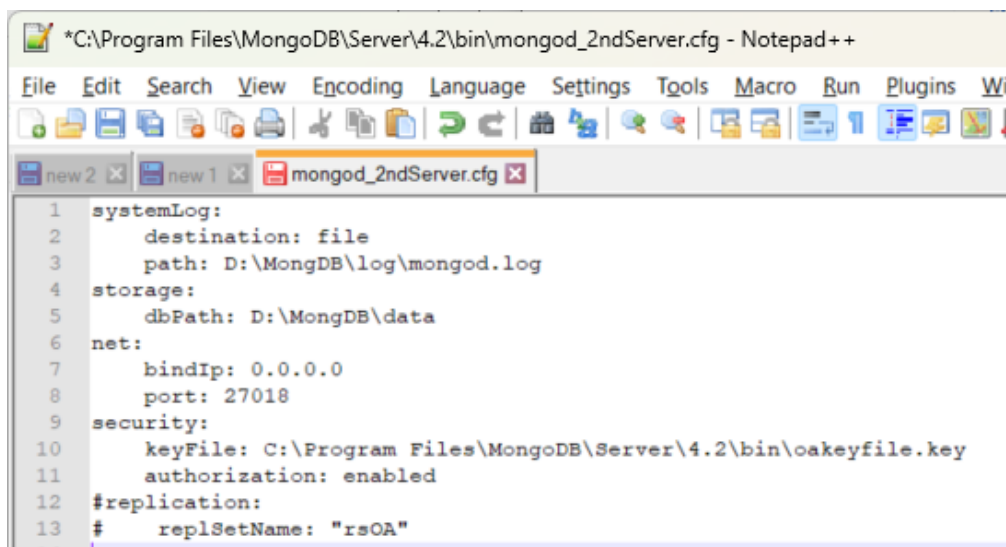
- Create a configuration file for your MongoDB instance by copy-pasting from the available ***mongod.cfg*** configuration file in the directory.

C:\Program Files\MongoDB\Server\4.2\bin

- Rename the new configuration file, e.g. **mongod_2ndServer.cfg**.
- Edit the information in the configuration file:

systemLog:Path	The path to the Database\log directory
storage:dbPath	The path to the Database\data directory

net:port	TCP port
replication:replSetName	Replication Deployment



```
*C:\Program Files\MongoDB\Server\4.2\bin\mongod_2ndServer.cfg - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Win
new 2 x new 1 x mongod_2ndServer.cfg x
1 systemLog:
2   destination: file
3   path: D:\MongoDB\log\mongod.log
4 storage:
5   dbPath: D:\MongoDB\data
6 net:
7   bindIp: 0.0.0.0
8   port: 27018
9 security:
10  keyFile: C:\Program Files\MongoDB\Server\4.2\bin\oakeyfile.key
11  authorization: enabled
12 #replication:
13 #   replSetName: "rsOA"
```

Step 3: Create a Service

- Open the Windows Power Shell app.
- Run the MongoDB Shell.

```
> cd C:\Program Files\MongoDB\Server\4.2\bin
> mongo
```

With MongoDB version 4., the MongoDB Shell call is > mongo

For MongoDB version 5., the MongoDB Shell call is > mongosh

- Run the Service installation commands corresponding to the created configuration file.

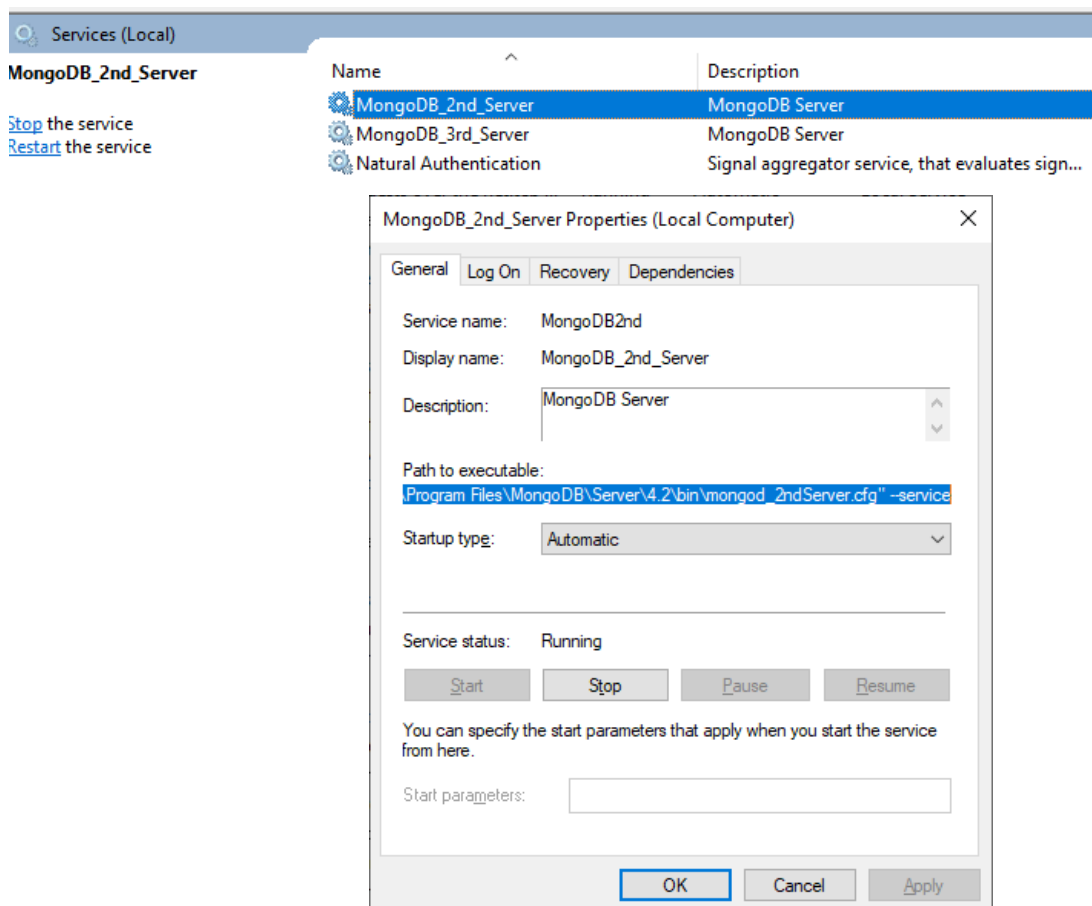
```
> mongod --config "C:\Program
Files\MongoDB\Server\4.2\bin\mongod_2ndServer.cfg"
--serviceName MongoDB2nd --serviceDisplayName
MongoDB_2nd_Server --install
```

In the example above

C:\Program Files\MongoDB\Server\4.2\bin\mongod_2ndServer.cfg	The path to the newly created configuration file for the MongoDB instance
serviceName MongoDB2nd	

serviceDisplayName MongoDB_2nd_Server	Service Information	Registration
--	------------------------	--------------

- Service Installation Successfully → No Error in Mongo Shell. Check the list of registered Window Services.



B.2.2.3 Standalone Deployment

Open the mongod.cfg file, using the "#" sign to note/invalidate the two replication lines, → replSetName #replication, #replSetName:

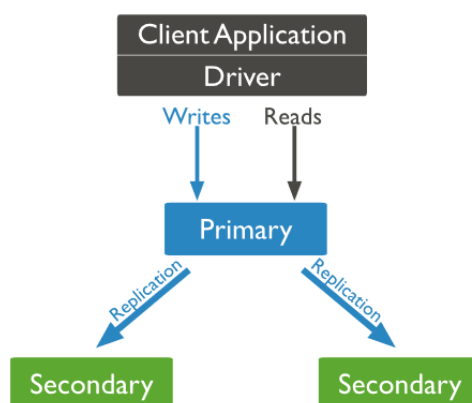
```
systemLog:
  destination: file
  path: C:\Program Files\MongoDB\Server\4.2\log\mongod.log
storage:
  dbPath: C:\Program Files\MongoDB\Server\4.2\data
net:
  bindIp: 0.0.0.0
  port: 27017
security:
  keyFile: C:\Program Files\MongoDB\Server\4.2\bin\oakeyfile.key
  authorization: enabled
#replication:
  #replSetName: "rsOA"
```

Restart the **MongoDB Server** service.

B.2.2.4 Replication Deployment

Replication is a way of synchronizing data through multiple servers. Provides a data redundancy mechanism to help prevent database loss on a single server, recover data in the event of a hardware failure or service interruption.

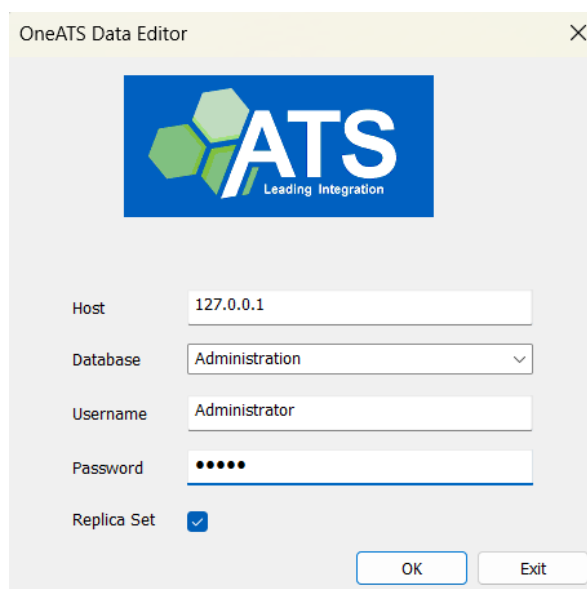
MongoDB deploys replication using a **replica set** – a group of instances containing the same dataset (the recommended number of instances is an odd number – optimizing hardware costs and ensuring data security).



First, it is necessary to ensure a good connection between the replication servers: for each server, perform a test by **telnet** to the other servers via port **27017**.

Refer to **the ChapterH.OneMasterTool/** configuration to enable **Replication**

After configuration, restart **the MongoDB Server** service on the servers.

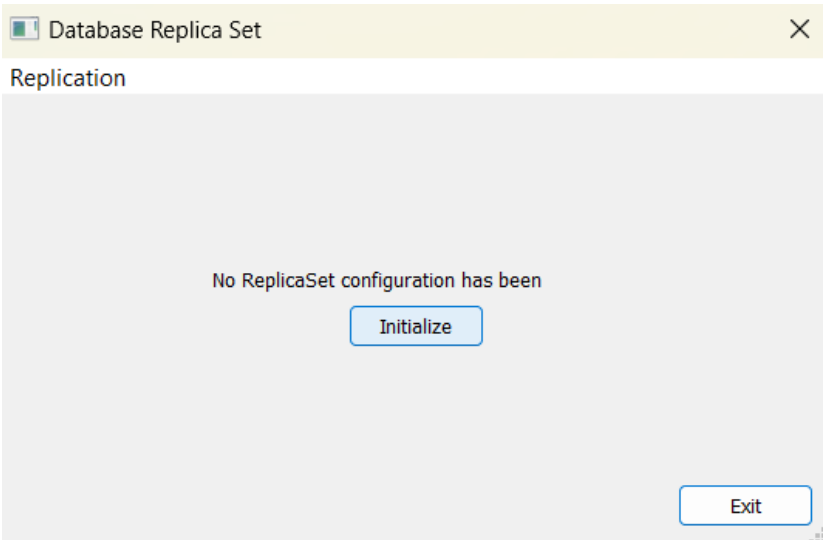


Start **the OneATS Data Editor** application, log in to Database Administration.

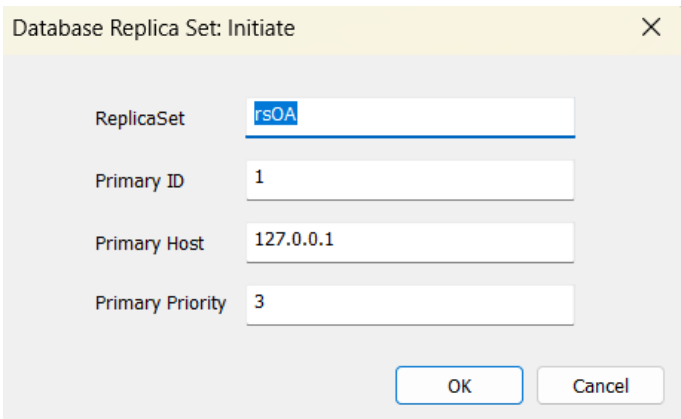
B.2.2.4.1 Replication Initilize

If MongoDB has not been initialized, the system will ask to initialize a Relica Set.

On the OneATS Data Editor app, log in to Database Administration, select File/Replication

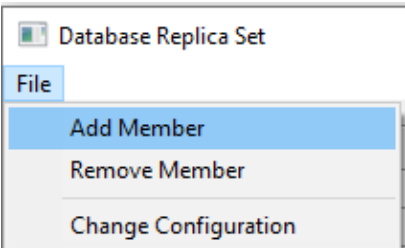


Select **Initialize**

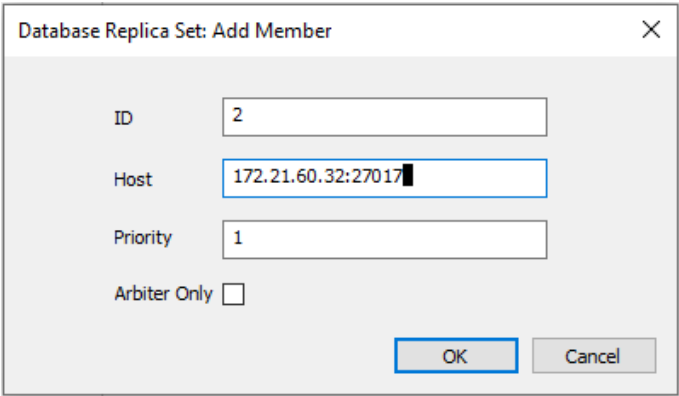


ReplicaSet	Replication Name
Primary ID	MongoDB node-ID primary
Primary Host	IP port of MongoDB node
Primary Priority	Priority - Give each member a distinct priority. - The member with the highest priority will be Primary.

B.2.2.4.2 Add Member



In the **Database Replica Set window**, select **File/Add Member** to add new members to the replica set.

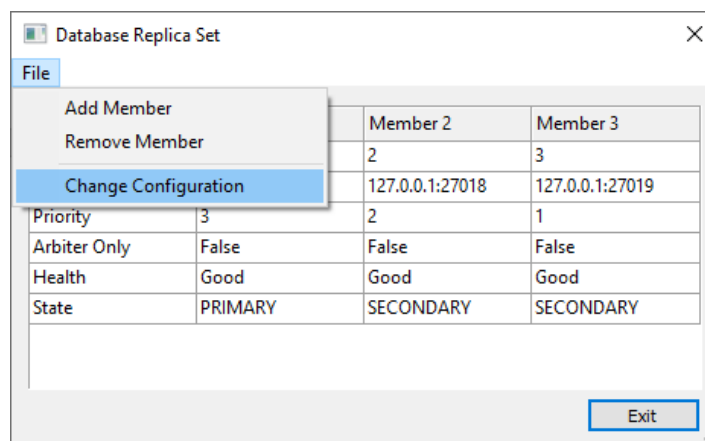


Fill in the information for the MongoDB node.

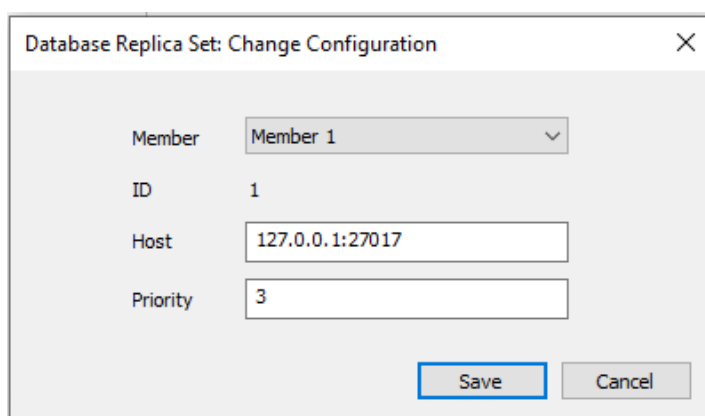
ID	The ID of the MongoDB Node.
Host	IP:Port of MongoDB Node
Priority	Priority - Give each member a distinct priority. - The member with the highest priority will be Primary.
Arbiter Only	Node Arbiter type. Arbiters are nodes that only participate in the voting process to determine the primary in the system, but do not contain \data

Select **OK**.

B.2.2.4.3 Edit Member Information

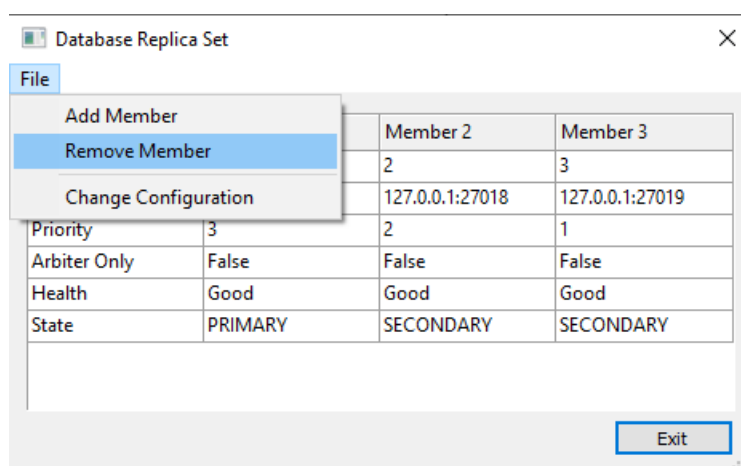


Users can adjust parameters such as **Host**, **Priority** of each member by selecting **File/Change Configuration**.



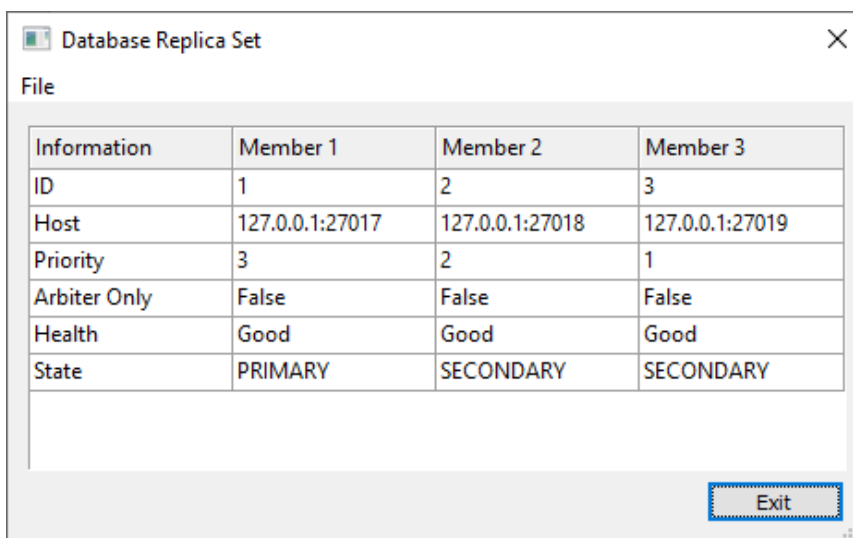
Select **Save** to save it.

B.2.2.4.4 Remove Member



In the **Database Replica Set** window, select **File/Remove Member** to add a new member to the replica set.

B.2.2.5 Replication Configuration & Status Check



The screenshot shows a window titled "Database Replica Set" with a "File" menu. It contains a table with 4 columns: "Information", "Member 1", "Member 2", and "Member 3". The table lists various attributes for three database members, including their IDs, hostnames, priorities, whether they are arbiters, their health status, and their current state (Primary or Secondary). An "Exit" button is located at the bottom right of the window.

Information	Member 1	Member 2	Member 3
ID	1	2	3
Host	127.0.0.1:27017	127.0.0.1:27018	127.0.0.1:27019
Priority	3	2	1
Arbiter Only	False	False	False
Health	Good	Good	Good
State	PRIMARY	SECONDARY	SECONDARY

Start the **OneATS Data Editor application**, log in to Database Replication.

The **Database Replica Set window** displays the configuration information and status of the members in Replication.



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